

Module 08: Planning in Robotics — Trajectory and Motion Planning

Learning Objectives

1. Define **trajectory planning** as the computation of a time-parameterized path from current state to goal state that satisfies physical constraints.
2. Analyze how **sampling-based planners (RRT, PRM) and optimization-based planners (trajectory optimization, MPC)** implement policy search over continuous state and action spaces.
3. Apply the EFE framework to understand the trade-off between optimality and computational tractability in robotic planning.

Introduction

Planning for robots operating in continuous state and action spaces requires finding feasible, safe, and efficient trajectories through high-dimensional configuration spaces. This is the most computationally demanding component of the Active Inference loop — evaluating the Expected Free Energy of candidate trajectories through complex, constrained spaces.

Key Concepts

1. Configuration Space and Obstacle Representations

Planning operates in **configuration space (C-space)** — the space of all possible robot configurations:

- A 6-DOF arm has a 6-dimensional C-space (6 joint angles)
- Obstacles in task space map to irregular regions in C-space — configurations where any part of the robot collides with an obstacle
- The planning problem: find a path through C-space from the start configuration to the goal configuration that avoids all obstacle regions

2. Sampling-Based Planning (RRT, PRM)

When C-space is too complex for analytical solutions, **sampling-based planners** explore by randomly sampling configurations:

- **Rapidly-exploring Random Trees (RRT)**: Grows a tree from the start configuration by randomly sampling new configurations and extending the tree toward them. RRT* adds rewiring to approach optimality.
- **Probabilistic Roadmaps (PRM)**: Pre-computes a graph of sampled configurations connected by collision-free edges. Planning queries find paths through this graph.
- Active Inference interpretation: Sampling is **epistemic exploration** of C-space — each sample reveals whether a configuration is feasible, progressively building a model of the free space.

3. Trajectory Optimization

Trajectory optimization computes smooth, dynamically feasible trajectories by solving a constrained optimization problem:

- **Cost function** = Expected Free Energy: a combination of goal deviation (pragmatic), control effort (energy cost), and constraint violation penalties (obstacle proximity, joint limits)
- **Constraints**: Joint limits, velocity limits, torque limits, collision avoidance, dynamic feasibility
- **Methods**: CHOMP (gradient descent on cost), TrajOpt (sequential convex optimization), iLQR (iterative linear-quadratic regulation)

This is direct EFE minimization in continuous state-action space — the planner searches for the trajectory that minimizes the sum of pragmatic cost and constraint violations.

4. Reactive vs. Deliberative Planning

Real robots combine both modes:

- **Deliberative planning** (offline or slow online): Compute an optimal trajectory to the goal — handles global structure (room connectivity, obstacle layout)

- **Reactive planning** (fast online): Adjust the trajectory in real time to handle dynamic changes (moving obstacles, unexpected contacts) — handles local perturbations
- Active Inference naturally integrates both through hierarchical planning — high-level deliberation sets the policy tree root, low-level reactive inference handles moment-to-moment adjustments

Applications

- **Robotic surgery path planning:** A surgical robot plans instrument trajectories through the patient's anatomy using trajectory optimization — minimizing tissue interaction forces while satisfying no-go zones around critical structures (nerves, blood vessels) specified in pre-operative imaging.
- **Autonomous vehicle route planning:** A self-driving car combines deliberative planning (route selection through the road network) with reactive planning (real-time trajectory adjustment for traffic, pedestrians, and road conditions) — a hierarchical Active Inference architecture operating at seconds-to-hours timescales.

Conclusion

Trajectory and motion planning implements Active Inference policy selection in continuous spaces — C-space exploration, trajectory optimization, and reactive-deliberative integration all minimize Expected Free Energy under physical constraints. This completes the Control and Estimation unit.